

Evaluation of EEG and MEG responses during Fine Motor Imagery from the same limb

Rui Jiang^{1,3,*}, Shuang Qiu^{1,2,3,*}, Yu Wang^{3,4}, Chuncheng Zhang^{1,2,†}, Huiguang He^{1,2,3}

¹ Laboratory of Brain Atlas and Brain-Inspired Intelligence, Chinese Academy of Sciences, Beijing, China

² Key Laboratory of Brain Cognition and Brain-inspired Intelligence Technology, Chinese Academy of Sciences, Beijing, China.

³ University of Chinese Academy of Sciences, Beijing, China.

⁴ National Engineering & Technology Research Center for ASIC Design, Chinese Academy of Sciences, Beijing, China.

Abstract—Motor imagery (MI) is an important brain-computer interface (BCI) paradigm. BCI systems based on fine MI can provide an intuitive control pathway of the outer device. Electroencephalography (EEG) is a widely used modality for MI due to its high temporal resolution and portability. Magnetoencephalography (MEG) has high spatial and temporal resolution, which has received more and more attention. This study designed four kinds of MI tasks of different joints from the same upper limb, including finger, wrist, elbow, and shoulder joints, and additionally added a resting task. The EEG and MEG signals of eight subjects were acquired synchronously. Analysis was conducted on the EEG and MEG data to find the time, time-frequency, and spatial difference between MI tasks of different joints from the same limb. The induced event-related desynchronization (ERD) in EEG signals at the electrode position of the left motor area are more broad and stronger in the alpha frequency band than that in MEG signals during fine MI tasks. From the topographical distribution, different MI tasks affects the area and intensity of the activated area, and topographical distribution of MEG signals in different MI tasks are more discriminative than that of EEG signals. Moreover, the analysis of movement-related cortical potentials (MRCP) showed that significant negative potentials were detected near the onset of the motor imagery events and there is a significant difference in temporal dimension between magnetoencephalogram and electroencephalogram signals. The work implies that there exist the separable differences between EEG and MEG during fine MI tasks, which can be utilized to build a multimodal classification method for fine MI-BCI systems.

Keywords—BCI, MI, MEG, EEG, Event-related spectral perturbation (ERSP), MRCP

I. INTRODUCTION

Brain-computer interface (BCI) systems capture and analyze brain signals and convert them into output commands, which provide a neuromuscular pathway that translates the brain activities of humans to commands for controlling devices or prostheses [1]. BCIs include some main experimental paradigms, such as P300 potentials, steady-state visual evoked potentials (SSVEPs), and motor imagery (MI). Compared

with other BCI paradigms, MI reflects the subject's voluntary movement awareness without any external stimulus [2].

Currently in the MI paradigm, imagined movements are defined as the imagining of kinesthetic movements of large body parts such as the hands, feet, and tongue, through which the imagining of these movements can modulate brain activity [3]. The simple Limb MI-BCI system only enables the control output of a single movement of different limbs, making it difficult to meet the demand for fine control of different joints of the same limb. Moreover, these modes often result in inconsistencies between the imagining task and the output patterns of the peripheral devices, which in turn lead to motor intention and cognitive disassociation between end effectors [4]. Therefore, research on different joint MI paradigms for the same limb is necessary.

Brain signals commonly used for non-invasive brain-computer interfaces include electroencephalogram (EEG), magnetoencephalography (MEG), functional magnetic resonance imaging (fMRI), and functional near-infrared spectroscopy (fNIRS). Among them, EEG has been used widely due to its high temporal resolution and portability. Meanwhile, MEG has gained more and more attention due to its high spatial and temporal resolution, because the complementary activations in the magnetic domain are considerably less distorted by the inhomogeneous conductivity of the head [5]. Moreover, MEG not only carries complementary information in terms of source depth and conductivity sensitivities, but also radially/tangentially oriented dipole detection [6]. However, none of these studies have explored the brain response of MEG in the MI paradigm, lacked comparisons of the two signals in different dimensions. There were no analyses based on the spatial location of brain activation, frequency bands, or leads.

In this study, we designed the paradigm of fine MI from the same limb and recorded bimodal data including EEG and MEG signals. Fine MI experiments include five tasks (hand, wrist, elbow, shoulder, and rest). Analysis was conducted on the EEG and MEG data to find the time, frequency, and spatial

This work was supported by the National Natural Science Foundation of China (U21A20388, 62276262) and by the Beijing Natural Science Foundation (7222311 and J210010).

*These authors contributed equally to this paper.

†Corresponding Author. Email: chuncheng.zhang@ia.ac.cn

difference between MI tasks of different joints from the same limb. It will be helpful to develop MEG-based fine MI-BCI systems and bimodal (MEG-EEG) fine MI-BCI.

II. MATERIALS AND METHODS

A. Participants.

Eight healthy participants who were right-handed (3 males and 5 females, 22-25 years old) participated in this experiment. All subjects signed an informed consent form before experiments. The study was approved by the Ethics Committee of the Institute of Automation, Chinese Academy of Sciences.

B. Experimental Paradigm

Experiments were designed and five categorical datasets were collected for the right side, finger, wrist, elbow, shoulder, and rest. The experiment consisted of a total of eight sessions. In each session, one category contained 8 trials, and there were 5 categories with a total of 40 trials displayed in random order. At the beginning of each trial, a red circle was displayed in the center of the screen for 2 seconds to remind subjects to pay more attention. After that, a text prompt ("hand", "wrist", "elbow", "shoulder", or "+", where "+" indicates a resting state) appears on the screen for a duration of 4s, subjects needed to follow the text displayed on the screen to imagine the corresponding limb movement or to This phase lasted for 4 s. a white circle was displayed in the center of the screen for 2 s, reminding the subjects that the trial was over. Until the start of the next trial, the red ball cue appeared on the screen again. The order in which the five types of motor imagery tasks appeared in a session was random to ensure the accuracy of the data obtained from the experiment.

C. Data Collection and Preprocessing

EEG and MEG signals were acquired simultaneously. Subjects were asked to perform the corresponding motor

imagery according to the prompts on the screen. During the motor imagery, subjects were asked to try not to blink and avoid any movement. Before the formal experiment, subjects were asked to perform two pre-training sessions.

In experiments conducting simultaneous MEG and EEG recordings, the same 275-channel CTF MEGTM system was used with a sampling rate of 200 Hz. Strict control was exerted over the participants' head movement, limiting it to a maximum of 5 millimeters. EEG data was recorded using 35 electrodes placed at standard scalp locations. The reference electrode (serving as a measurement baseline) was positioned at the vertex, while the ground electrode was placed at the AFz position. To minimize EEG interference in MEG recordings, the impedance of each EEG electrode was strictly controlled to be below 5 k Ω .

The acquired EEG and MEG data were preprocessed. The preprocessing steps were based on MATLAB 2019a software, using and EEGLAB toolkit (v14.1.0b) and Fieldtrip toolkit. Data with large errors as well as clearly visible anomalies, such as those outside the recording range or with consecutive zero amplitudes, were deleted and complemented with surrounding channels. The signal is de-baselined and down sampled to 200 Hz to reduce computational effort. The downsampled data is band-pass filtered from 1 to 40 Hz to filter out high frequency and low frequency noise.

D. Event-related spectral perturbation

Event-related spectral perturbation (ERSP) was first proposed by Makeig in 1993 [7] and measures average dynamic changes in amplitude of the broad band EEG frequency spectrum as a function of time relative to an experimental event [8]. ERSP calculates the power spectrum of the data in each trial over a sliding time window centered on time t and then averages over all trials. The average ERSP at time t with frequency f is defined as follows:

$$ERSP(f, t) = \frac{1}{n} \sum_{k=1}^n |F_k(f, t)|^2 \quad (1)$$

where n denotes the number of trials and $F_k(f, t)$ is the power density of the k^{th} trial at time point t and frequency f .

To analyze the similarities and differences between EEG and EEG signals in the time-frequency domain, we performed ERSP using the fast Fourier transform method in the range of 1~40 Hz. five types of average time-frequency maps were plotted on the C3, Cz, and C4 leads.

Brain topology maps can display the power values in a certain frequency band of a brain signal in different colors on a head-table plan, which facilitates quantitative analysis of brain signals in the time and frequency domains [9]. It is effective to see in which areas ERDs induced by different cognitive tasks occur.

E. Movement-related Cortical Potential

Movement-related cortical potential (MRCP) signals have been proved to detect different types of movements and

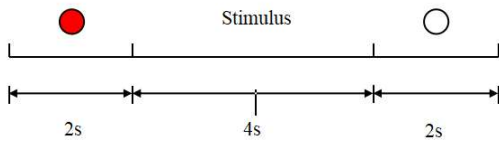


Fig. 1. Timing of one trial of our experimental paradigm. The experiment consists of 8 sessions in total. Each session contains 40 trials, and the duration of each trial is 8s.

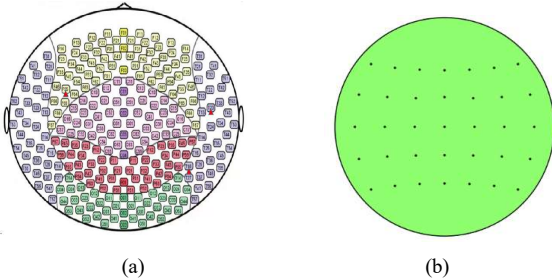


Fig. 2. Distribution of MEG (a) and EEG (b) electrode and probe positions.

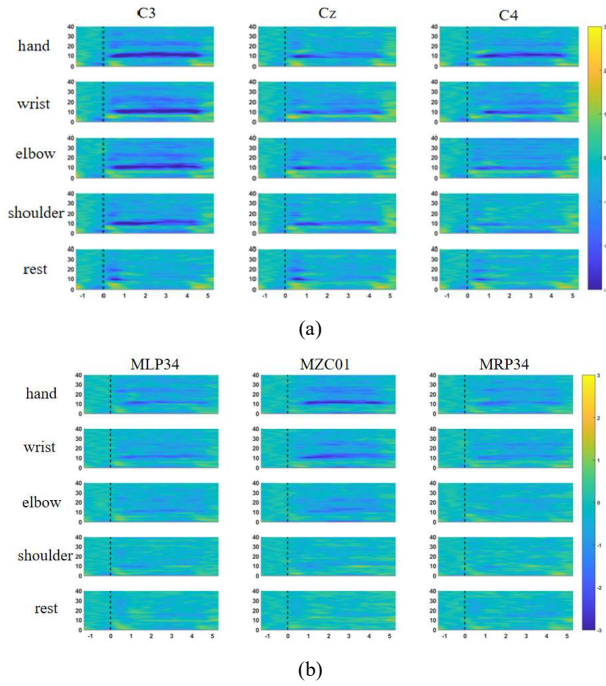


Fig. 3. Averaged time-frequency maps of eight subject of EEG(a) and MEG (b). blue indicates ERD.

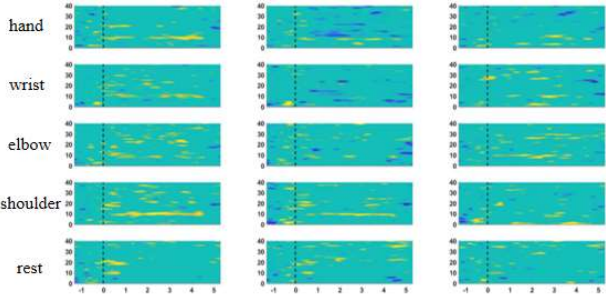


Fig. 4. Comparison of ERSP between EEG and MEG (yellow and blue areas indicate $p < 0.05$).

distinguish between rest and motion. They reflect many neurophysiological processes and contain enough information to be associated with both intent and executed movements [10]. The MRCP and ERD features during MI tasks contain significantly discriminative information, which are complementary to each other [11]. Thus, we performed MRCP analysis is contributing to a more comprehensive knowledge of the brain signals evoked by motor imagery.

If a warning stimulus is provided followed by a cue to start the movement the MRCP is generally termed the contingent negative variation (CNV) [12]. The CNV consists of several components related to specific neurophysiological mechanisms. The first deflection, the early CNV, typically commences immediately following the warning stimulus [13]. Specifically, the CNV only occurs preceding movement/imagery and subsides immediately upon movement/imagery execution [3].

III. RESULT AND Discussion

A. ERSP Analysis

Fig. 3 shows the average time-frequency maps of all eight subjects. The ERD/ERS phenomenon can be seen (the blue part of the figure indicates the ERD phenomenon) in both Fig. 3(a) and Fig. 3(b), especially in the alpha band with a time of 0.5~4.0 seconds and a frequency of around 10~12 Hz. For both EEG and MEG, clear contralateral dominance of the ERD phenomenon for the four MI categories of finger, wrist, elbow, and shoulder can be seen, whereas there is almost no obvious ERD phenomenon in the resting state. ERD was stronger in the fingers and wrists relative to the shoulder joint. The ERD of EEG was most obvious in the C3 channel in the left motor area, and the ERD of MEG was most obvious in the MCZ01 channel in the center motor area. Moreover, the ERDs of MEG appeared later and weaker than those of EEG.

According to the time-frequency maps (Fig. 3), we performed t-test analysis for comparison of ERSP between EEG and MEG, which has been shown in Fig. 4. We did t-test ($p^* \leq 0.05$) with ERSP values for each time and frequency for the eight subjects. The points where there was no significant difference ($p^* > 0.05$) were set zero. For points with significant differences, if the ERSP value of EEG was greater than that of MEG, the value was set to $(p - 1) \times 5$ (colored in blue); Otherwise, it was set to $(1 - p) \times 5$ (colored in yellow). In the channels of left motor area, ERSP differences were mainly seen in the alpha band. The ERD intensity of the EEG was stronger than that of the MEG during the MI task in both the left motor area and the right motor area. Comparing the Cz and MCZ01 channels revealed that, except for shoulder-related tasks, the MEG exhibits lower values of ERSP than EEG at specific time and frequency locations for other tasks.

B. Topographic ERD/ERS map

To further analyze the spatial distribution characteristics, we identified the most prominent ERD regions from the time-frequency plots. Then selected the 0-4s period and the alpha frequency band, and calculated the average ERSP values of all subjects within this region. By locating the frequency point corresponding to the minimum ERSP value and extracting a frequency range of 0.5Hz above and below this point, we generated the brain topographic maps shown in Fig. 5. Both EEG and MEG brain topography maps showed an obvious contralateral dominant ERD phenomenon except in the resting state. The activation degree of the finger, wrist, and elbow joints was relatively large, and the activation ranges varied. The activation was strongest in the wrist for EEG and in the finger for MEG. However, the activation gradually decreased in all tasks, with the shoulder joint having the lowest activation. This proved that the different joints of unilateral limbs had a certain degree of differentiation. Compared to EEG, the ERD distribution of MEG was more spatially focused and the overall level of ERD was weaker.

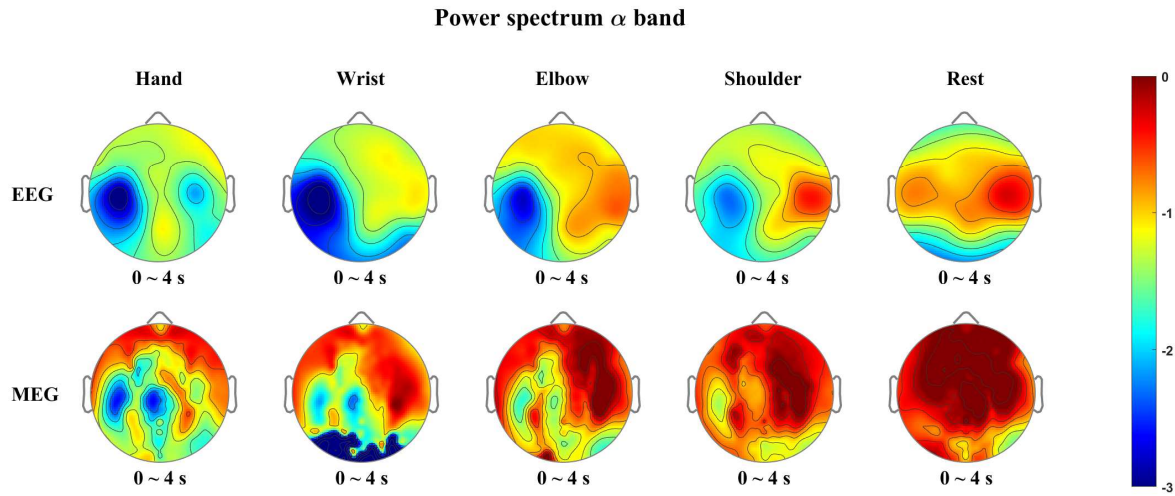


Fig. 5. Topographical distribution of power of EEG and MEG in the upper alpha band (10-12Hz), blue indicates ERD.

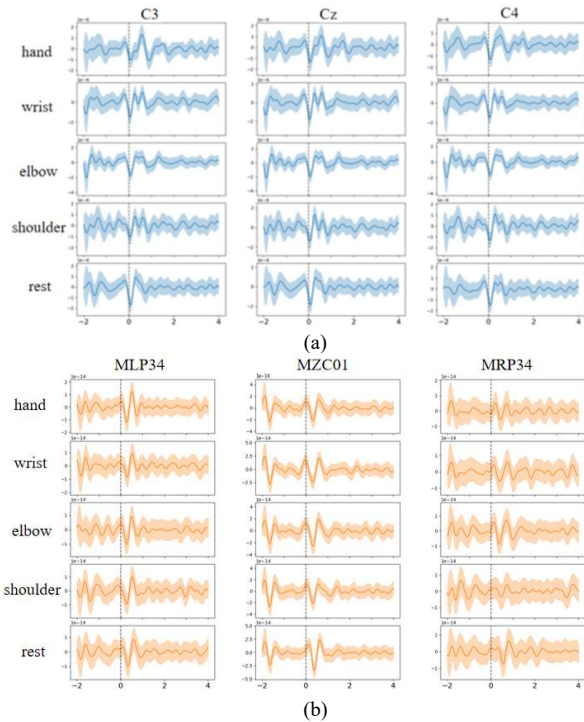


Fig. 6. Averaged MRCP Morphology of eight subjects of EEG (a) and MEG (b).

The different levels of activation in different joints may be due to the fact that different joints are applied with different frequency and dexterity in life. In other words, subjects are more familiar with joint movements of the fingers and wrist compared to the elbow and shoulder tasks, thus subjects are more familiar with MI tasks of the fingers and wrist compared to the elbow and shoulder MI tasks, which results in stronger brain activity in MI tasks of the fingers and wrist. Two areas of activation can be seen in the MEG brain topography of the fingers and wrist, in the left motor area of the brain and in the center. This corresponds to the apparent ERD phenomenon in

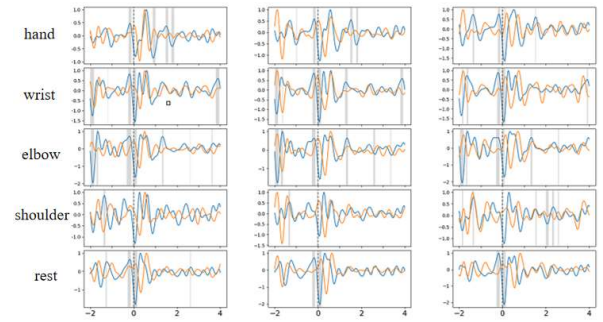


Fig. 7. Comparison of MRCP between EEG and MEG (grey areas indicate significant differences ($p < 0.01$)).

the alpha frequency band in the time-frequency maps for the finger and wrist at the MCZ01 electrodes. In the MEG of the brain ground map for the wrist task, there was an area of high activation in the occipital region, which is uncommon because the ERD phenomenon basically occurs in the contralateral motor area. Therefore, we checked and returned to the data acquisition and processing process to confirm that there was no error, and found that it was the activation in the occipital region of the 6th subject that led to the abnormal results after averaging, which may be due to individual differences.

C. MRCP

Brain signals were filtered between 1Hz to 3Hz to obtain MRCP. MRCP was calculated by averaging all trials for each of three typical channels for all participants.

Fig. 6(a) and Fig. 6(b) show average MRCP from all subjects for EEG and MEG. It can be seen that both EEG and MEG have obvious CNV at the onset of the MI tasks. The CNV phenomenon of EEG is delayed by 0.2~0.5 seconds compared with that of MEG, and the maximal CNV amplitude of EEG and MEG is observed in the Cz and MZC01 leads, respectively.

We also did t-test significance test ($p^* < 0.01$) on MRCP of EEG and MEG for different motor imagery tasks, and the results are shown in Fig .7. The large differences between EEG and MEG were mainly concentrated near the onset of the MI tasks. We also found that the difference in the shoulder is the smallest, which may be because the shoulder is the most difficult to imagine the movement so the effect on both EEG and MEG is not very good, resulting in these two do not see a significant difference.

Combined with the time-frequency diagrams, the ERD phenomenon was least obvious at the Cz electrode of EEG and the MZC01 electrode of MEG. We can find that MRCP and ERD exhibit different activation characteristics. The MRCP results in the current study are not in accordance with the ERSP analysis. This may be related to the different origins of ERD and MRCP reported in the literature. The source of MRCP is related to the cerebellar-thalamus-cortical pathway, while ERD is related to the thalamo-cortical feedback loops [14].

IV. CONCLUSION

We designed the paradigm of fine MI from the same limb and recorded bimodal data including EEG and MEG signals. Fine MI experiments include five tasks (hand, wrist, elbow, shoulder, and rest). Analysis was conducted on the EEG and MEG data to find the time, frequency, and spatial difference between MI tasks of different joints from the same limb. The induced ERD in EEG signals at the electrode position of the left motor area are more broad and stronger in the alpha frequency band than that in MEG signals during fine MI tasks. From the topographical distribution, different MI tasks affects the area and intensity of the activated area, and topographical distribution of MEG signals in different MI tasks are more discriminative than that of EEG signals. Moreover, the analysis of MRCP showed that significant negative potentials were detected near the onset of the MI events and there is a significant difference in temporal dimension between MEG and EEG signals. The work implies that there exist the separable differences between EEG and MEG during fine MI tasks, which can be utilized to build a multimodal classification method for fine MI-BCI systems.

REFERENCES

- [1] Y.-H. Liu, L.-F. Lin, C.-W. Chou, Y. Chang, Y.-T. Hsiao, and W.-C. Hsu, "Analysis of Electroencephalography Event-Related Desynchronisation and Synchronisation Induced by Lower-Limb Stepping Motor Imagery," *Journal of Medical and Biological Engineering*, vol. 39, 03/30, 2018.
- [2] G. Pfurtscheller, and C. Neuper, "Motor imagery and direct brain-computer communication," *Proceedings of the IEEE*, vol. 89, no. 7, pp. 1123-1134, 2001.
- [3] V. Morash, O. Bai, S. Furlani, P. Lin, and M. Hallett, "Classifying EEG signals preceding right hand, left hand, tongue, and right foot movements and motor imageries," *Clinical Neurophysiology*, vol. 119, no. 11, pp. 2570-2578, 2008/11/01/, 2008.
- [4] B. Edelman Bj Fau - Baxter, B. Baxter B Fau - He, and B. He, "EEG Source Imaging Enhances the Decoding of Complex Right-Hand Motor Imagery Tasks," no. 1558-2531 (Electronic).
- [5] B. Wittevrongel, N. Holmes, E. Boto, R. Hill, M. Rea, A. Libert, E. Khachatryan, M. Van Hulle, R. Bowtell, and M. Brookes, "Practical

- real-time MEG-based neural interfacing with optically pumped magnetometers," *BMC Biology*, vol. 19, 08/10, 2021.
- [6] M.-C. Corsi, M. Chavez, D. Schwartz, L. Hugueville, A. Khambhati, D. Bassett, and F. Fallani, "Integrating EEG and MEG Signals to Improve Motor Imagery Classification in Brain-Computer Interface," *International Journal of Neural Systems*, vol. 29, 11/20, 2017.
- [7] X. Ma, S. Qiu, W. Wei, S. Wang, and H. He, "Deep Channel-Correlation Network for Motor Imagery Decoding From the Same Limb," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 28, no. 1, pp. 297-306, 2020.
- [8] G. P. H. L. D. J. C. N. Silva, "Event-related EEG/MEG synchronization and desynchronization: basic principles," 1999.
- [9] B. Susan Philip, G. Prasad, and D. J. Hemanth, "A systematic review on artifact removal and classification techniques for enhanced MEG-based BCI systems," *Brain-Computer Interfaces*, vol. 10, no. 2-4, pp. 99-113, 2023/10/02, 2023.
- [10] R. R. Said, M. B. B. Heyat, K. Song, C. Tian, and Z. Wu, "A Systematic Review of Virtual Reality and Robot Therapy as Recent Rehabilitation Technologies Using EEG-Brain–Computer Interface Based on Movement-Related Cortical Potentials," vol. 12, no. 12, pp. 1134, 2022.
- [11] K. Wang, M. Xu, Y. Wang, S. Zhang, L. Chen, and D. Ming, "Enhance decoding of pre-movement EEG patterns for brain-computer interfaces," *Journal of Neural Engineering*, vol. 17, no. 1, pp. 016033, 2020/01/24, 2020.
- [12] N. Mrachacz-Kersting, S. Kristensen, I. Niazi, and D. Farina, "Precise temporal association between cortical potentials evoked by motor imagination and afference induces cortical plasticity," *The Journal of physiology*, vol. 590, pp. 1669-82, 01/16, 2012.
- [13] T. Hamano, H. O. Lüders, A. Ikeda, T. F. Collura, Y. G. Comair, and H. Shibasaki, "The cortical generators of the contingent negative variation in humans: a study with subdural electrodes," *Electroencephalography and Clinical Neurophysiology/Evoked Potentials Section*, vol. 104, no. 3, pp. 257-268, 1997/05/01/, 1997.
- [14] R. Xu, N. Jiang, A. Vuckovic, M. Hasan, N. Mrachacz-Kersting, D. Allan, M. Fraser, B. Nasserolelami, B. Conway, K. Dremstrup, and D. Farina, "Movement-related cortical potentials in paraplegic patients: Abnormal patterns and considerations for BCI-rehabilitation," *Frontiers in Neuroengineering*, vol. 7, 08/27, 2014.